

Locally Thermal from Globally Athermal: Correlations as a Thermodynamic Resource under Gibbs-Preserving Operations

Alexander Belik

Supervisor: Felix Binder

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Abstract

We study a symmetric bipartite quantum system AA' at inverse temperature β with identical local Hamiltonians $H_A = H_{A'} =: H$ and local Gibbs state $\gamma = e^{-\beta H}/Z$. A bipartite state $\rho_{AA'}$ is called *locally thermal* (LT) if both marginals are exactly thermal:

$$\mathrm{Tr}_{A'} \rho_{AA'} = \gamma, \quad \mathrm{Tr}_A \rho_{AA'} = \gamma.$$

This removes all local athermality, so any deviation from the global equilibrium state $\Gamma := \gamma \otimes \gamma$ is stored entirely in correlations. The project asks three linked questions: what is the structure of the LT set, how correlations evolve under global Gibbs-preserving operations, and how the convertibility hierarchy changes when one restricts to local Gibbs-preserving maps.

The main analytic contribution of the project is the explicit low-dimensional characterization of LT states, with the full two-qubit problem as the principal target. In particular, we show that LT is an affine slice of the positive semidefinite cone, derive its exact dimension, obtain the full 9-parameter two-qubit LT normal form, and reduce generic two-qubit LT positivity to an explicit Schur-complement semi-algebraic system. We also analyse controlled lower-dimensional subfamilies, including the diagonal- T slice and ray constructions, which serve as analytic and numerical warmups for the full body. In dimension $d = 3$, where the full LT set is 64-dimensional and not analytically tractable in complete detail, we develop the commuting qutrit LT subclass as a standalone higher-dimensional subproject and identify it with a 4-dimensional transportation polytope.

On the operational side, we prove that on the LT manifold the free-energy monotone $D(\rho\|\Gamma)$ coincides exactly with the mutual information $I(A:A')_\rho$, establish closure of LT under local Gibbs-preserving operations, and present an explicit restricted-family convertibility theorem on the thermofield-double-to-dephased chain. Finally, we formulate semidefinite programs (SDPs) for LT membership, projection to LT, and global and local Gibbs-preserving convertibility, and we indicate where the numerical results generated by the codebase fit into the final report.

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1 Introduction

1.1 Problem setting and motivation

Modern thermodynamics increasingly treats information as physical. Correlations, uncertainty, and coherence affect the energetic value of states, and the possibility of work extraction. In the quantum regime this becomes especially rich because correlations may be classical, coherent, or genuinely entangled. The central theme of this project is a particularly clean setting where all local nonequilibrium features have been removed, so that the only possible thermodynamic resource is stored globally in correlations.

We consider a symmetric bipartite system AA' with

$$H_A = H_{A'} =: H, \quad \gamma = \frac{e^{-\beta H}}{Z}, \quad \Gamma := \gamma \otimes \gamma.$$

A state $\rho_{AA'}$ is *locally thermal* if both reduced states are exactly equal to the local Gibbs state γ . Such a state may still differ from the product thermal state Γ , but this difference can no longer be attributed to local athermality. It is therefore natural to interpret the remaining deviation as a resource encoded purely in correlations.

The guiding research questions are:

1. What is the structure of bipartite quantum states with thermal marginals?
2. How do correlations in such states evolve under global Gibbs-preserving operations?
3. How does the convertibility structure change when the allowed operations are restricted to local Gibbs-preserving maps?

1.2 Project scope and narrative

The cleanest way to organise the project is by separating three levels of difficulty.

1. **Main analytic target: the full two-qubit LT body.** For $d = 2$, the LT set has real dimension $(2^2 - 1)^2 = 9$. This is the main analytic goal of the project: to characterise the full 9-dimensional two-qubit LT set as explicitly as possible, not merely a few low-dimensional slices.
2. **Warmup and supporting analysis: slices and rays.** Before attacking the full 9-dimensional body, it is useful to study controlled subfamilies such as the diagonal- T slice and one-parameter LT rays. These are not the main theorem of the project, but they give exact analytic tests, intuition about geometry, and numerically stable probes of convertibility.

3. **Higher-dimensional exploration: qutrit subclasses.** For $d = 3$, the full LT set already has real dimension $(3^2 - 1)^2 = 64$, which is too large for a complete analytic characterization in the same style. The realistic goal is therefore not solve the full qutrit LT problem but rather identify analytically meaningful subclasses. The commuting qutrit LT family provides exactly such a nontrivial case.

This hierarchy is what gives the project its internal logic:

general LT geometry $\rightarrow$ full qubit characterization $\rightarrow$ slice/ray warmups and convertibility probes

1.3 Summary of results

The report establishes the following analytic results.

1. The LT set is an affine slice of the positive semidefinite cone and is therefore a spectrahedron.
2. Its real dimension is

$$\dim_{\mathbb{R}} \text{LT}(\gamma_A, \gamma_B) = (d_A^2 - 1)(d_B^2 - 1).$$

3. For symmetric qubits, every LT state admits an explicit 9-parameter matrix normal form.
4. Generic positivity for the full two-qubit LT family reduces to an explicit Schur-complement criterion, giving a semi-algebraic description of the full body.
5. On the LT manifold,

$$D(\rho \parallel \Gamma) = I(A : A')_{\rho},$$

so global athermality relative to equilibrium equals total correlations.

6. LT is closed under local Gibbs-preserving maps.
7. The diagonal- T family admits exact positivity and separability criteria.
8. The commuting qutrit LT subclass is a 4-dimensional transportation polytope with exact mutual-information range

$$0 \leq I \leq H(g).$$

9. The thermofield-double-to-dephased family forms an explicitly ordered chain under local Gibbs-preserving operations.

2 Background

2.1 Quantum states, channels, and correlations

A finite-dimensional quantum state is a density operator $\rho \in \mathcal{D}(\mathcal{H})$ satisfying

$$\rho = \rho^\dagger, \quad \rho \succeq 0, \quad \text{Tr } \rho = 1.$$

For a bipartite state ρ_{AB} , the marginals are

$$\rho_A = \text{Tr}_B \rho_{AB}, \quad \rho_B = \text{Tr}_A \rho_{AB}.$$

Physical transformations are modelled by completely positive trace-preserving (CPTP) maps, also called quantum channels. We use Kraus, Stinespring, and Choi representations as needed. In particular, the Choi formalism is the natural language for the SDPs used later for convertibility.

Correlation classes include:

- **product states:** $\rho_{AB} = \rho_A \otimes \rho_B$;
- **separable states:** $\rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}$;
- **entangled states:** states that are not separable.

2.2 Entropy, relative entropy, and mutual information

The von Neumann entropy is

$$S(\rho) := -\text{Tr}(\rho \log \rho).$$

The quantum relative entropy is

$$D(\rho \parallel \sigma) := \text{Tr}[\rho(\log \rho - \log \sigma)],$$

defined whenever $\text{supp}(\rho) \subseteq \text{supp}(\sigma)$.

For a bipartite state ρ_{AB} , the mutual information is

$$I(A:B)_\rho = S(\rho_A) + S(\rho_B) - S(\rho_{AB}),$$

and admits the relative-entropy representation

$$I(A:B)_\rho = D(\rho_{AB} \parallel \rho_A \otimes \rho_B).$$

A central structural fact is data processing:

$$D(\rho\|\sigma) \geq D(\Phi(\rho)\|\Phi(\sigma))$$

for every CPTP map Φ .

2.3 Gibbs states, free energy, and entropy production

Given a Hamiltonian H and inverse temperature β , the Gibbs state is

$$\gamma = \frac{e^{-\beta H}}{Z}, \quad Z := \text{Tr}(e^{-\beta H}).$$

The Helmholtz free energy is

$$F(\rho) = \text{Tr}(\rho H) - \beta^{-1} S(\rho).$$

Relative entropy to the Gibbs state has direct thermodynamic meaning:

$$D(\rho\|\gamma) = \beta(F(\rho) - F(\gamma)).$$

Thus relative entropy to thermal equilibrium quantifies nonequilibrium free-energy content.

For a process exchanging heat Q with a bath, the entropy production is

$$\sigma := \Delta S - \beta Q \geq 0.$$

2.4 Gibbs-preserving operations

Definition 2.1 (Gibbs-preserving channel). Let γ be a Gibbs state. A CPTP map Φ is *Gibbs-preserving* (GP) if

$$\Phi(\gamma) = \gamma.$$

Proposition 2.2 (GP iff free-energy monotonicity for all inputs). *Let Φ be CPTP and let γ be a Gibbs state. Then*

$$\forall \rho : \quad D(\rho\|\gamma) \geq D(\Phi(\rho)\|\gamma) \quad \Longleftrightarrow \quad \Phi(\gamma) = \gamma.$$

Proof. If $\Phi(\gamma) = \gamma$, the claim follows from data processing with reference state γ . Conversely, if the inequality holds for all ρ , substitute $\rho = \gamma$ to obtain

$$0 = D(\gamma\|\gamma) \geq D(\Phi(\gamma)\|\gamma) \geq 0,$$

hence $\Phi(\gamma) = \gamma$. □

3 Locally Thermal States

3.1 Definition and affine decomposition

Definition 3.1 (Locally thermal state). For fixed local Gibbs states γ_A and γ_B , define

$$\text{LT}(\gamma_A, \gamma_B) := \left\{ \rho_{AB} \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B) : \text{Tr}_B \rho_{AB} = \gamma_A, \text{Tr}_A \rho_{AB} = \gamma_B \right\}.$$

In the symmetric case $\gamma_A = \gamma_B = \gamma$, we write simply $\text{LT}(\gamma)$.

Proposition 3.2 (Affine-slice characterization). *A state ρ_{AB} is locally thermal if and only if*

$$\rho_{AB} = \Gamma + C, \quad C = C^\dagger, \quad \text{Tr}_A C = 0, \quad \text{Tr}_B C = 0, \quad \Gamma + C \succeq 0,$$

where $\Gamma := \gamma_A \otimes \gamma_B$. Equivalently,

$$\text{LT}(\gamma_A, \gamma_B) = (\Gamma + \mathcal{Z}) \cap \text{PSD},$$

where

$$\mathcal{Z} := \{C = C^\dagger : \text{Tr}_A C = 0, \text{Tr}_B C = 0\}$$

is a real linear subspace.

Proof. If $\rho \in \text{LT}(\gamma_A, \gamma_B)$, define $C := \rho - \Gamma$. Then C is Hermitian, has zero partial traces, and $\Gamma + C = \rho \succeq 0$. Conversely, any such C gives a positive semidefinite operator with the required marginals. □

Remark 3.3. This shows immediately that LT is a *spectrahedron*: an affine slice of the positive semidefinite cone. In particular it is convex and closed.

3.2 Dimension counting

Proposition 3.4 (Dimension of the LT set). *Let $d_A = \dim \mathcal{H}_A$ and $d_B = \dim \mathcal{H}_B$. Then*

$$\dim_{\mathbb{R}} \text{LT}(\gamma_A, \gamma_B) = (d_A^2 - 1)(d_B^2 - 1).$$

In particular, for symmetric $d \times d$ systems,

$$\dim_{\mathbb{R}} \text{LT}(\gamma) = (d^2 - 1)^2.$$

Proof. A bipartite density operator on $\mathcal{H}_A \otimes \mathcal{H}_B$ has $(d_A d_B)^2 - 1$ real degrees of freedom. Fixing the A marginal imposes $d_A^2 - 1$ independent real linear constraints, and fixing the B marginal imposes $d_B^2 - 1$ more. Therefore

$$(d_A d_B)^2 - 1 - (d_A^2 - 1) - (d_B^2 - 1) = (d_A^2 - 1)(d_B^2 - 1).$$

□

For qubits ($d = 2$), this gives

$$\dim_{\mathbb{R}} \text{LT} = (2^2 - 1)^2 = 9,$$

which is why the full two-qubit LT body is a genuine 9-dimensional problem rather than a three-parameter toy model. For qutrits ($d = 3$), the dimension is

$$\dim_{\mathbb{R}} \text{LT} = (3^2 - 1)^2 = 64,$$

which already explains why the qutrit part of the project must be more modest.

3.3 Correlation tensor parameterization

Let $\{F_\mu\}_{\mu=0}^{d_A^2-1}$ and $\{G_\nu\}_{\nu=0}^{d_B^2-1}$ be orthonormal Hermitian operator bases with

$$F_0 = \frac{\mathbb{I}_A}{\sqrt{d_A}}, \quad G_0 = \frac{\mathbb{I}_B}{\sqrt{d_B}},$$

and F_μ, G_ν traceless for $\mu, \nu \geq 1$. Then every zero-marginal Hermitian operator $C \in \mathcal{Z}$ has expansion

$$C = \sum_{\mu=1}^{d_A^2-1} \sum_{\nu=1}^{d_B^2-1} c_{\mu\nu} F_\mu \otimes G_\nu, \quad c_{\mu\nu} \in \mathbb{R}.$$

Hence LT is parameterized by a real $(d_A^2 - 1) \times (d_B^2 - 1)$ correlation tensor, with positivity of $\Gamma + C$ as the only nonlinear constraint.

4 Core Thermodynamic Structure on LT

4.1 TFD state and pure LT uniqueness

Let

$$\gamma = \sum_i g_i |i\rangle\langle i|$$

be diagonal in the energy basis. The thermofield-double vector is

$$|\tilde{\gamma}\rangle = \sum_i \sqrt{g_i} |i\rangle_A \otimes |i\rangle_{A'}.$$

Its density matrix

$$\tilde{\gamma}_{AA'} := |\tilde{\gamma}\rangle\langle\tilde{\gamma}|$$

is pure and satisfies

$$\mathrm{Tr}_{A'} \tilde{\gamma}_{AA'} = \gamma = \mathrm{Tr}_A \tilde{\gamma}_{AA'}.$$

Proposition 4.1 (Uniqueness of pure LT states). *If $\rho_{AA'} = |\psi\rangle\langle\psi|$ is pure and satisfies*

$$\mathrm{Tr}_{A'} \rho_{AA'} = \gamma = \mathrm{Tr}_A \rho_{AA'},$$

then $|\psi\rangle$ is a TFD state up to local unitaries acting inside degenerate eigenspaces of γ . In the nondegenerate case,

$$|\psi\rangle = \sum_i \sqrt{g_i} e^{i\theta_i} |i\rangle \otimes |i\rangle,$$

so the only freedom is local phase choice.

Proof sketch. Any pure bipartite state admits a Schmidt decomposition. The reduced state fixes the Schmidt coefficients to be $\{\sqrt{g_i}\}$. If the spectrum of γ is nondegenerate, the Schmidt basis is fixed by the energy basis up to phases. Degeneracies enlarge the freedom to local unitaries inside the degenerate blocks. $\square$

4.2 Athermality equals correlations on LT

This identity is conceptually central: within the LT manifold, thermodynamic resource relative to Γ is exactly total correlation.

Proposition 4.2 (On LT, $D(\rho\|\Gamma) = I(A:B)_\rho$). *If $\rho \in \mathrm{LT}(\gamma_A, \gamma_B)$, then*

$$D(\rho\|\gamma_A \otimes \gamma_B) = I(A:B)_\rho.$$

In the symmetric case,

$$D(\rho\|\Gamma) = I(A:A')_\rho.$$

Proof. Use the general identity

$$I(A:B)_\rho = D(\rho\|\rho_A \otimes \rho_B),$$

and then substitute $\rho_A = \gamma_A$ and $\rho_B = \gamma_B$ on LT. $\square$

Remark 4.3. This equality is the bridge between thermodynamic and informational language in the whole project. On LT, “free-energy gap above equilibrium” and “total correlations” are not merely related; they are exactly the same quantity.

4.3 Closure under local Gibbs-preserving operations

Proposition 4.4 (Closure of LT under local GP maps). *Let Φ_A and Φ_B be channels such that*

$$\Phi_A(\gamma_A) = \gamma_A, \quad \Phi_B(\gamma_B) = \gamma_B.$$

Then for any $\rho \in \text{LT}(\gamma_A, \gamma_B)$,

$$(\Phi_A \otimes \Phi_B)(\rho) \in \text{LT}(\gamma_A, \gamma_B).$$

Proof. By linearity of partial trace,

$$\text{Tr}_B \left[(\Phi_A \otimes \Phi_B)(\rho) \right] = \Phi_A(\text{Tr}_B \rho) = \Phi_A(\gamma_A) = \gamma_A,$$

and similarly for the other marginal. □

This proves that local GP operations cannot concentrate global athermality into local marginals. A global GP map may still leave the LT set, but a local GP map cannot.

5 Result: Full Two-Qubit LT Characterization

5.1 Why the qubit case is the main analytic target

For symmetric qubits, the LT set has dimension 9. This is the first nontrivial case where one can hope to obtain a near-complete analytic description of the full LT body, rather than only of a few low-dimensional sections. This is therefore the natural place where the project can move from “examples” to “structure theorem”.

Throughout this section let

$$\gamma = \begin{pmatrix} g & 0 \\ 0 & 1 - g \end{pmatrix}, \quad \frac{1}{2} \leq g < 1.$$

The nondegenerate case $g \neq 1/2$ is the thermodynamically biased regime; the infinite-temperature case $g = 1/2$ will be discussed separately as a special simplification.

5.2 Exact canonical matrix form

Proposition 5.1 (Canonical 9-parameter two-qubit LT form). *A two-qubit state ρ is locally thermal with both marginals equal to γ if and only if it can be written in the product energy basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ as*

$$\rho = \begin{pmatrix} g - a & x & y & z \\ x^* & a & w & -y \\ y^* & w^* & a & -x \\ z^* & -y^* & -x^* & 1 - g - a \end{pmatrix}, \quad a \in \mathbb{R}, \quad x, y, z, w \in \mathbb{C}, \quad (1)$$

with the sole remaining condition $\rho \succeq 0$.

Proof. Begin with a general 4×4 Hermitian density matrix. Imposing

$$\mathrm{Tr}_{A'} \rho = \gamma, \quad \mathrm{Tr}_A \rho = \gamma$$

gives six independent real linear constraints, reducing the original 15 real degrees of freedom to 9. Solving these constraints explicitly yields (1). $\square$

Remark 5.2. This is the exact full two-qubit LT family. It is already enough to show why the project should not be described as “just analysing slices”: the true problem is the positivity geometry of (1).

5.3 Immediate necessary bounds from principal minors

Positivity of diagonal entries and 2×2 principal minors implies

$$0 \leq a \leq \min(g, 1 - g), \quad g - a \geq 0, \quad 1 - g - a \geq 0,$$

and

$$|x|^2 \leq a(g - a), \quad |y|^2 \leq a(g - a), \quad |w|^2 \leq a(1 - g - a), \quad |z|^2 \leq (g - a)(1 - g - a).$$

These inequalities are useful but not sufficient: they do not capture the coupled constraints among the five free parameters (a, x, y, z, w) .

5.4 Local GP unitary gauge

The local unitaries that preserve a nondegenerate qubit Gibbs state are exactly diagonal phase rotations in the energy basis:

$$U_A = \text{diag}(e^{i\phi_0}, e^{i\phi_1}), \quad U_B = \text{diag}(e^{i\psi_0}, e^{i\psi_1}).$$

Conjugating ρ by $U_A \otimes U_B$ preserves the LT constraints and preserves positivity. Under this action,

$$x \mapsto e^{i(\psi_0 - \psi_1)} x, \quad y \mapsto e^{i(\phi_0 - \phi_1)} y,$$

so without loss of generality one may choose phases so that

$$x \geq 0, \quad y \geq 0$$

are both real.

Define

$$\alpha := g - a, \quad \delta := 1 - g - a.$$

A convenient gauge-fixed normal form is therefore

$$\rho = \begin{pmatrix} \alpha & x & y & z \\ x & a & w & -y \\ y & w^* & a & -x \\ z^* & -y & -x & \delta \end{pmatrix}, \quad x, y \in \mathbb{R}_{\geq 0}, \quad z, w \in \mathbb{C}. \quad (2)$$

5.5 Schur-complement reduction of the full positivity problem

This is the key analytic theorem for the qubit part of the project.

Partition (2) into 2×2 blocks,

$$\rho = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}, \quad A = \begin{pmatrix} \alpha & x \\ x & a \end{pmatrix}, \quad B = \begin{pmatrix} y & z \\ w & -y \end{pmatrix}, \quad D = \begin{pmatrix} a & -x \\ -x & \delta \end{pmatrix}.$$

Let

$$\Delta := \det A = a\alpha - x^2.$$

Theorem 5.3 (Generic positivity criterion for two-qubit LT states). *Let ρ be a gauge-fixed LT state of the form (2). Assume $\Delta > 0$. Then*

$$\rho \succeq 0 \iff \begin{cases} 0 \leq a \leq 1 - g, \\ \Delta \geq 0, \\ N_{11} \geq 0, \quad N_{22} \geq 0, \quad N_{11}N_{22} \geq |N_{12}|^2, \end{cases}$$

where

$$N := D - B^\dagger A^{-1} B = \begin{pmatrix} N_{11} & N_{12} \\ N_{12}^* & N_{22} \end{pmatrix},$$

and

$$A^{-1} = \frac{1}{\Delta} \begin{pmatrix} a & -x \\ -x & \alpha \end{pmatrix}.$$

The explicit entries are

$$N_{11} = a - \frac{ay^2 + \alpha|w|^2 - 2xy \Re(w)}{\Delta}, \quad (3)$$

$$N_{22} = \delta - \frac{\alpha y^2 + a|z|^2 + 2xy \Re(z)}{\Delta}, \quad (4)$$

$$N_{12} = \frac{-x\Delta - ayz + \alpha yw + xzw - xy^2}{\Delta}. \quad (5)$$

Proof. For $\Delta > 0$, the principal block A is invertible. The block Schur-complement criterion states

$$\rho \succeq 0 \iff A \succeq 0 \text{ and } D - B^\dagger A^{-1} B \succeq 0.$$

Substituting the explicit matrices gives the formulas above. Since N is 2×2 Hermitian,

$$N \succeq 0 \iff N_{11} \geq 0, \quad N_{22} \geq 0, \quad N_{11}N_{22} \geq |N_{12}|^2.$$

□

Remark 5.4. This theorem gives the explicit semi-algebraic description of the *generic* interior of the full qubit LT body. It is the strongest analytic result in the project because it reduces the 9-dimensional positivity problem to concrete inequalities.

5.6 Boundary case $\Delta = 0$

The rank-drop boundary $\Delta = 0$ is not physically exceptional; it is the boundary of the principal block A . In that case one should use the pseudoinverse form of the Schur criterion:

$$\rho \succeq 0 \iff A \succeq 0, \quad (\mathbb{I} - AA^+)B = 0, \quad D - B^\dagger A^+ B \succeq 0.$$

Equivalently, positivity on the boundary is the continuous closure of the generic inequalities as $\Delta \downarrow 0$, together with the range condition

$$\text{Ran}(B) \subseteq \text{Ran}(A).$$

For the main body of the report it is enough to state this boundary criterion and reserve the rank-one algebra for an appendix.

6 Warmup Families: Slices, Rays, and Internal Structure

6.1 Why these are warmups rather than the main theorem

The project does not reduce to slices. However, slices are still useful because they give:

- (a) exact feasibility inequalities,
- (b) exact separability criteria in tractable subfamilies,
- (c) visually interpretable geometry,
- (d) controlled numerical testbeds for convertibility SDPs.

For that reason they belong in the report, but as supporting analysis around the main qubit characterization, not as a replacement for it.

6.2 Classical diagonal qubit LT line

Restricting to commuting LT states in the qubit case gives

$$\rho_{\text{cl}} = \text{diag}(q, g - q, g - q, 1 - 2g + q),$$

where fixed marginals force

$$\max(0, 2g - 1) \leq q \leq g.$$

Equivalently, the diagonal coupling matrix

$$P = \begin{pmatrix} q & g - q \\ g - q & 1 - 2g + q \end{pmatrix}$$

is exactly the 2×2 transportation polytope with margins $(g, 1 - g)$.

This line runs from the product Gibbs point to the extremal classical coupling and is the simplest LT subfamily. It is useful as a baseline but does not probe quantum coherence.

6.3 Pauli correlation tensor and the full 9-dimensional qubit parameterization

Let

$$z_\gamma := 2g - 1.$$

Every two-qubit LT state can be written as

$$\rho = \frac{1}{4} \left(\mathbb{I} \otimes \mathbb{I} + z_\gamma (\sigma_z \otimes \mathbb{I} + \mathbb{I} \otimes \sigma_z) + \sum_{i,j \in \{x,y,z\}} T_{ij} \sigma_i \otimes \sigma_j \right), \quad (6)$$

with $T \in \mathbb{R}^{3 \times 3}$.

This representation makes the dimension count transparent: the LT set is the set of all real 3×3 tensors T such that $\rho(T) \succeq 0$.

6.4 The diagonal- T slice

A particularly useful three-parameter slice is obtained by restricting

$$T = \text{diag}(t_x, t_y, t_z).$$

Then

$$\rho = \Gamma + \frac{1}{4} (t_x \sigma_x \otimes \sigma_x + t_y \sigma_y \otimes \sigma_y + t_z \sigma_z \otimes \sigma_z).$$

In the product energy basis this is an X -state:

$$\rho = \begin{pmatrix} g^2 + \frac{t_z}{4} & 0 & 0 & \frac{t_x - t_y}{4} \\ 0 & g(1 - g) - \frac{t_z}{4} & \frac{t_x + t_y}{4} & 0 \\ 0 & \frac{t_x + t_y}{4} & g(1 - g) - \frac{t_z}{4} & 0 \\ \frac{t_x - t_y}{4} & 0 & 0 & (1 - g)^2 + \frac{t_z}{4} \end{pmatrix}.$$

Define

$$a := g^2 + \frac{t_z}{4}, \quad b := g(1 - g) - \frac{t_z}{4}, \quad d := (1 - g)^2 + \frac{t_z}{4},$$

and

$$u := \frac{t_x - t_y}{4}, \quad v := \frac{t_x + t_y}{4}.$$

Then positivity is equivalent to positivity of two 2×2 blocks:

$$\begin{pmatrix} a & u \\ u & d \end{pmatrix} \succeq 0, \quad \begin{pmatrix} b & v \\ v & b \end{pmatrix} \succeq 0.$$

Hence the diagonal- T state is feasible if and only if

$$a \geq 0, \quad d \geq 0, \quad ad \geq u^2, \tag{7}$$

$$b \geq 0, \quad b^2 \geq v^2. \tag{8}$$

Equivalently,

$$g^2 + \frac{t_z}{4} \geq 0, \tag{9}$$

$$(1 - g)^2 + \frac{t_z}{4} \geq 0, \tag{10}$$

$$(4g^2 + t_z)(4(1 - g)^2 + t_z) \geq (t_x - t_y)^2, \tag{11}$$

$$t_z + |t_x + t_y| \leq 4g(1 - g). \tag{12}$$

6.5 Infinite-temperature limit

At $\beta = 0$ one has $g = \frac{1}{2}$ and $\gamma = \mathbb{I}/2$. Then the diagonal- T conditions simplify to

$$1 - t_z \geq |t_x + t_y|, \quad 1 + t_z \geq |t_x - t_y|,$$

equivalently

$$1 \pm t_x \pm t_y \pm t_z \geq 0$$

for all sign choices. This is exactly the Bell-diagonal tetrahedron.

This limit is useful because it gives a familiar benchmark. At finite temperature the same

family persists but the tetrahedron deforms according to the thermal bias.

6.6 Exact separability criterion on the diagonal- T slice

In 2×2 dimensions, PPT is equivalent to separability. The partial transpose swaps the two off-diagonal couplings, giving

$$\rho^{T_B} = \begin{pmatrix} g^2 + \frac{t_z}{4} & 0 & 0 & \frac{t_x+t_y}{4} \\ 0 & g(1-g) - \frac{t_z}{4} & \frac{t_x-t_y}{4} & 0 \\ 0 & \frac{t_x-t_y}{4} & g(1-g) - \frac{t_z}{4} & 0 \\ \frac{t_x+t_y}{4} & 0 & 0 & (1-g)^2 + \frac{t_z}{4} \end{pmatrix}.$$

Thus $\rho^{T_B} \succeq 0$ is equivalent to

$$t_z + |t_x - t_y| \leq 4g(1-g), \quad (13)$$

$$(4g^2 + t_z)(4(1-g)^2 + t_z) \geq (t_x + t_y)^2. \quad (14)$$

Theorem 6.1 (Exact separability criterion on the diagonal- T slice). *A diagonal- T LT state is separable if and only if it satisfies the LT positivity inequalities and the PPT inequalities above. Equivalently,*

$$\rho(t_x, t_y, t_z) \in \text{Sep} \cap \text{LT}$$

if and only if

$$t_z + |t_x + t_y| \leq 4g(1-g), \quad (15)$$

$$t_z + |t_x - t_y| \leq 4g(1-g), \quad (16)$$

$$(4g^2 + t_z)(4(1-g)^2 + t_z) \geq (t_x + t_y)^2, \quad (17)$$

$$(4g^2 + t_z)(4(1-g)^2 + t_z) \geq (t_x - t_y)^2. \quad (18)$$

This gives an exact internal hierarchy on the diagonal- T family:

$$\text{classical LT} \subset \text{separable LT} \subset \text{LT}.$$

At infinite temperature these inequalities reduce further to

$$|t_x| + |t_y| + |t_z| \leq 1,$$

so the separable subset is the Bell-diagonal octahedron inside the Bell-diagonal tetrahedron.

6.7 Exact positivity bounds on LT rays

Given any Hermitian zero-marginal direction $C_0 \in \mathcal{Z}$, consider the LT ray

$$\rho(p) = \Gamma + pC_0.$$

Define the whitened operator

$$\tilde{C}_0 := \Gamma^{-1/2} C_0 \Gamma^{-1/2}, \quad \tilde{\rho}(p) := \Gamma^{-1/2} \rho(p) \Gamma^{-1/2} = \mathbb{I} + p\tilde{C}_0.$$

Since congruence preserves positivity,

$$\rho(p) \succeq 0 \iff \mathbb{I} + p\tilde{C}_0 \succeq 0.$$

If $\{\lambda_k\}$ are the eigenvalues of $\tilde{C}_0$, then the exact feasible interval is

$$1 + p\lambda_k \geq 0 \quad \forall k,$$

or equivalently

$$p \in \left[\max_{\lambda_k > 0} \left(-\frac{1}{\lambda_k} \right), \min_{\lambda_k < 0} \left(-\frac{1}{\lambda_k} \right) \right].$$

This ray analysis is useful for two reasons. First, it provides exact one-parameter feasibility bounds. Second, it improves numerical conditioning by factoring out the thermal background before scanning outward from Γ .

7 Qutrits: Full LT Geometry vs Tractable Subclasses

7.1 Why the full qutrit problem is not the main analytic theorem

For $d = 3$, the symmetric LT set has real dimension

$$(3^2 - 1)^2 = 64.$$

Analytically, the full qutrit LT body is therefore far larger than the qubit case. It is still useful to state its exact structure, but not realistic to claim a full explicit characterization in the same sense as for qubits. The correct project framing is:

The qubit problem is the main full characterization problem; the qutrit problem is explored through analytically meaningful subclasses.

7.2 Full qutrit LT as a 64-dimensional spectrahedron

Let $\{B_a\}_{a=1}^8$ be an orthonormal traceless Hermitian basis for qutrit operators, for example normalized Gell–Mann matrices. Then any symmetric qutrit LT state has the form

$$\rho = \Gamma + \sum_{a=1}^8 \sum_{b=1}^8 c_{ab} B_a \otimes B_b, \quad c \in \mathbb{R}^{8 \times 8},$$

with the sole nonlinear constraint $\rho \succeq 0$.

Thus the full qutrit LT set is a 64-dimensional spectrahedron. This exact statement is important, but it does not yet produce a practical analytic classification.

7.3 Commuting qutrit LT as a transportation polytope

The most natural structured qutrit subclass is the commuting family. Let

$$\gamma = \text{diag}(g_1, g_2, g_3), \quad g_i > 0, \quad g_1 + g_2 + g_3 = 1.$$

A commuting qutrit LT state is equivalent to a nonnegative matrix $P = (p_{ij})_{i,j=1}^3$ with row and column sums both equal to g :

$$\sum_j p_{ij} = g_i, \quad \sum_i p_{ij} = g_j.$$

The corresponding density matrix is

$$\rho_P = \sum_{i,j=1}^3 p_{ij} |ij\rangle \langle ij|.$$

This is exactly the transportation polytope $\mathcal{T}(g, g)$.

A convenient explicit parameterization is

$$P = \begin{pmatrix} a & b & g_1 - a - b \\ c & d & g_2 - c - d \\ g_1 - a - c & g_2 - b - d & g_3 - g_1 + a + b + c + d \end{pmatrix},$$

subject to entrywise nonnegativity. This makes the dimension 4 manifest.

7.4 Exact mutual-information range in the commuting subclass

For commuting states the mutual information reduces to the classical mutual information of the coupling P :

$$I(P) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{g_i g_j}.$$

Equivalently,

$$I(P) = 2H(g) - H(P),$$

where

$$H(g) = - \sum_i g_i \log g_i, \quad H(P) = - \sum_{i,j} p_{ij} \log p_{ij}.$$

Proposition 7.1 (Exact correlation range on the commuting qutrit face). *For the commuting qutrit LT subclass,*

$$0 \leq I(P) \leq H(g).$$

The minimum $I = 0$ is attained uniquely at the product coupling

$$P_{\text{prod}} = gg^T.$$

The maximum $I = H(g)$ is attained at deterministic permutation couplings

$$P_{ij}^* = g_i \delta_{i,\pi(j)}$$

compatible with the equal-marginal constraint.

Proof. The lower bound follows from nonnegativity of relative entropy:

$$I(P) = D(P \| gg^T) \geq 0,$$

with equality if and only if $P = gg^T$. For the upper bound,

$$I(X : Y) = H(X) - H(X|Y) \leq H(X) = H(g),$$

with equality if and only if X is a deterministic function of Y . Since both marginals equal g , equality requires a deterministic bijective coupling, i.e. a permutation coupling. $\square$

7.5 Support graphs, face lattice, and vertices

The commuting qutrit LT polytope is not just numerically convenient; it has a clean exact combinatorial structure.

For a support set $S \subseteq \{1, 2, 3\} \times \{1, 2, 3\}$, let G_S be the bipartite support graph inside

$K_{3,3}$. If $c(S)$ denotes the number of connected components of G_S , then the corresponding face F_S has dimension

$$\dim F_S = |S| - 6 + c(S).$$

Hence:

- codimension-one faces correspond to support patterns with one forced zero;
- edges correspond to $\dim F_S = 1$;
- vertices correspond to $\dim F_S = 0$.

Proposition 7.2 (Vertex criterion). *A feasible coupling $P \in \mathcal{T}(g, g)$ is a vertex if and only if its support graph contains no cycle. Equivalently, vertices are exactly the feasible spanning forests of $K_{3,3}$.*

In the generic nondegenerate case every vertex has exactly $3 + 3 - 1 = 5$ positive entries, so each vertex support is a spanning tree on the six row/column vertices.

This is enough to justify a dedicated qutrit analytic subsection even before any numerical SDP results are inserted:

$$\text{exact dimension } 4 \quad + \quad \text{exact range } 0 \leq I \leq H(g) \quad + \quad \text{exact vertex/face structure.}$$

8 Restricted-Family Convertibility Theorem

8.1 The TFD-to-dephased chain

Let

$$|\tilde{\gamma}\rangle = \sum_i \sqrt{g_i} |ii\rangle, \quad \tilde{\gamma} := |\tilde{\gamma}\rangle\langle\tilde{\gamma}|,$$

and let Δ denote full product-energy dephasing:

$$\Delta(\rho) = \sum_{i,j} (|i\rangle\langle i| \otimes |j\rangle\langle j|) \rho (|i\rangle\langle i| \otimes |j\rangle\langle j|).$$

Then

$$\Delta(\tilde{\gamma}) = \sum_i g_i |ii\rangle\langle ii| =: \tilde{\gamma}_{\text{diag}}.$$

Dephasing is Gibbs-preserving because it fixes every diagonal Gibbs state. It also preserves local thermality because it does not change diagonal marginals.

Define the interpolation family

$$\rho_\lambda := (1 - \lambda)\tilde{\gamma} + \lambda\tilde{\gamma}_{\text{diag}}, \quad 0 \leq \lambda \leq 1.$$

Every ρ_λ lies in LT.

8.2 An explicit local GP map

Define

$$G_\mu := (1 - \mu) \text{id} + \mu \Delta, \quad 0 \leq \mu \leq 1.$$

Since both id and Δ are CPTP and both fix Γ , the map G_μ is global GP. Since $\Delta = \Delta_A \otimes \Delta_B$ and each local dephasing is Gibbs-preserving, G_μ is in fact allowed under local GP as well.

Applying G_μ to ρ_λ gives

$$G_\mu(\rho_\lambda) = (1 - \mu)\rho_\lambda + \mu\tilde{\gamma}_{\text{diag}} = (1 - \lambda')\tilde{\gamma} + \lambda'\tilde{\gamma}_{\text{diag}},$$

where

$$\lambda' = \lambda + \mu(1 - \lambda).$$

Theorem 8.1 (Exact ordered chain on the TFD-to-dephased family). *For $0 \leq \lambda \leq \lambda' \leq 1$,*

$$\rho_\lambda \succ_{\text{LGP}} \rho_{\lambda'} \quad \text{and hence} \quad \rho_\lambda \succ_{\text{GP}} \rho_{\lambda'}.$$

Indeed, choosing

$$\mu = \frac{\lambda' - \lambda}{1 - \lambda}$$

gives

$$G_\mu(\rho_\lambda) = \rho_{\lambda'}.$$

Proof. Direct substitution. □

Because each ρ_λ is LT,

$$D(\rho_\lambda \| \Gamma) = I(\rho_\lambda).$$

By data processing under G_μ ,

$$I(\rho_\lambda) \geq I(\rho_{\lambda'}) \quad (\lambda \leq \lambda').$$

Thus this family gives an explicit example where one has:

analytic state family $\rightarrow$ explicit free map $\rightarrow$ strict monotone ordering.

This is much stronger than an existential convertibility statement and should be highlighted as a report-quality theorem.

9 Local GP Action on Centered Correlations

Define centered correlations by

$$C := \rho - \Gamma, \quad \text{Tr}_A C = \text{Tr}_B C = 0.$$

For qubits, represent a local channel Φ by an affine Bloch map

$$r \mapsto Mr + t.$$

If $\Phi(\gamma) = \gamma$, then

$$t = (\mathbb{I} - M)r_\gamma, \quad r_\gamma = (0, 0, 2g - 1).$$

On centered correlations the translations cancel, so local GP acts linearly:

$$C \mapsto (\Phi_A \otimes \Phi_B)(C).$$

In the Pauli correlation tensor representation this becomes

$$T \mapsto T' = M_A T M_B^T.$$

This motivates singular values of T , or more invariantly operator-Schmidt singular values of C , as natural candidates for necessary local-convertibility diagnostics. They are not expected to be sufficient, but they are analytically meaningful and numerically accessible.

10 Computational Methods

10.1 Why SDPs are the natural numerical tool

The project is structured so that the analytically defined LT geometry feeds directly into convex optimization. LT membership is positivity plus linear marginal constraints, which is exactly a semidefinite feasibility problem. Convertibility under Gibbs-preserving channels can be expressed in the Choi picture by linear constraints on the Choi matrix together with positivity and trace-preservation. This makes semidefinite programming the natural numerical framework.

10.2 SDP 1: LT membership

Given a candidate bipartite state ρ , LT membership asks whether

$$\rho \succeq 0, \quad \text{Tr}_A \rho = \gamma, \quad \text{Tr}_B \rho = \gamma, \quad \text{Tr} \rho = 1.$$

This is a feasibility SDP.

10.3 SDP 2: Projection to LT

To quantify distance from LT one may solve

$$\min_{\sigma \in \text{LT}(\gamma)} \frac{1}{2} \|\rho - \sigma\|_1,$$

or an equivalent SDP formulation of trace-distance projection. This is useful for projecting randomly sampled states to the LT manifold before convertibility tests.

10.4 SDP 3: Global GP convertibility

Given LT states ρ and σ , global GP convertibility asks whether there exists a CPTP map Φ with Choi matrix J_Φ such that

$$\Phi(\rho) = \sigma, \quad \Phi(\Gamma) = \Gamma.$$

In Choi form these become linear constraints on J_Φ , together with

$$J_\Phi \succeq 0, \quad \text{Tr}_{\text{out}} J_\Phi = \mathbb{I}_{\text{in}}.$$

10.5 SDP 4: Local GP convertibility

Local convertibility asks whether there exist local channels Φ_A, Φ_B satisfying

$$\Phi_A(\gamma) = \gamma, \quad \Phi_B(\gamma) = \gamma, \quad (\Phi_A \otimes \Phi_B)(\rho) = \sigma.$$

This is more difficult than the global problem because of the tensor-product structure. The code uses verified local tests and also compares against convex outer relaxations such as PPT constraints on joint Choi objects.

10.6 Numerical conventions used in the code

For any figure or table intended for the report, local convertibility should be generated with

```
local_edge_mode = verified,
```

not heuristic single-start mode. The core report-facing experiments are:

```
lt_geometry_combined,  
lt_family_diagT_validation,  
lt_convertibility_graph,  
local_gp_ppt_relax,  
tfd_vs_dephased,  
separable_vs_entangled_lt.
```

Each run writes a deterministic results folder containing configuration and summary files together with figures and array outputs.

10.7 Numerical confirmation of the LT identity $D(\rho\|\Gamma) = I(A:B)$

A central analytic result of the project is that on the locally thermal manifold,

$$D(\rho\|\Gamma) = I(A:B)_\rho, \quad \Gamma = \gamma \otimes \gamma.$$

The numerics support this identity very strongly across all tested families.

First, in the benchmark comparison between the thermofield-double (TFD) state and its dephased version, the two quantities coincide exactly to numerical precision:

$$I(\text{TFD}) = 1.16441, \quad D(\text{TFD}) = 1.16441,$$

and

$$I(\text{deph}) = 0.582203, \quad D(\text{deph}) = 0.582203.$$

Second, along the one-parameter LT rays studied in the diagonal- T and Pauli- XX families, the plots of D and I lie on top of one another throughout the entire feasible interval. Third, in the $d = 3$ commuting LT subclass, the sampled mean values also agree:

$$\bar{I} = 0.0635303, \quad \bar{D} = 0.0635303,$$

with identical reported standard deviation

$$\text{std}(I) = \text{std}(D) = 0.0595621.$$

This repeated agreement across analytically distinct LT families is important for the overall narrative of the project. It shows that once the local marginals are fixed to γ , the nonequilibrium content relative to the product Gibbs state is entirely correlation-theoretic. Numerically, the thermodynamic and informational viewpoints become indistinguishable on LT, exactly as predicted analytically.

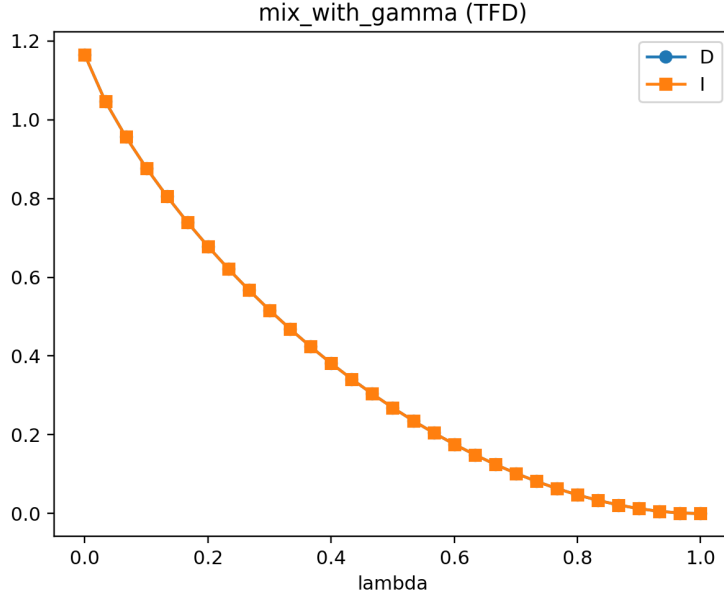


Figure 1: Thermalisation path obtained by mixing the TFD state with the product Gibbs state. Along this LT path, the relative entropy to equilibrium and the mutual information coincide throughout and decrease monotonically to zero as the state approaches $\Gamma = \gamma \otimes \gamma$.

10.8 PPT classification inside the sampled LT set

To probe the internal structure of the two-qubit LT body, random LT samples were classified by positivity under partial transpose. Since the system is 2×2 , PPT is equivalent to separability. Out of 80 sampled LT states, the PPT fraction was

$$0.113,$$

so under this sampling scheme only about 11.3% of sampled LT states were separable.

The scatter plot of sampled LT states in the (D, I) plane is especially informative. All points lie on the diagonal line

$$D = I,$$

again confirming the LT identity, but the PPT points cluster near the low-correlation region while the non-PPT points dominate as one moves outward from the product Gibbs state. In other words, for the sampling rule used here, weakly correlated LT states are often separable, while strongly correlated LT states are typically entangled.

This should be phrased carefully in the report. The figure does *not* prove that only a small geometric volume of LT is PPT in a measure-independent sense; it only shows that under the present random LT sampler, PPT states are relatively uncommon and are concentrated near the centre of the LT body. That is still a useful operational statement, because it supports the idea that fixing both marginals to thermal equilibrium does not eliminate entanglement. On the contrary, entanglement remains typical once one moves sufficiently far from the product Gibbs point inside LT.

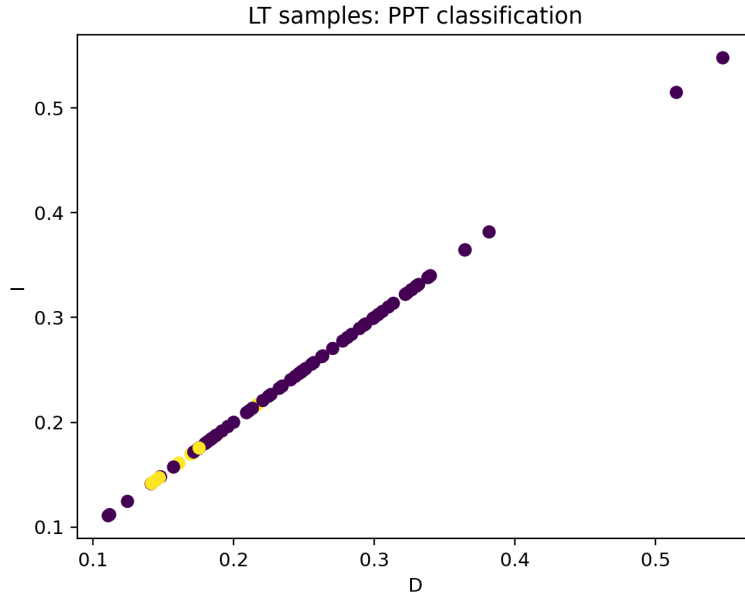


Figure 2: PPT classification of sampled two-qubit LT states in the (D, I) plane. All states lie on the diagonal $D = I$, while the PPT samples are concentrated near the low-correlation region. This indicates that under the present LT sampling rule, separable LT states occur mainly near the product Gibbs point, whereas more strongly correlated LT states are typically entangled.

10.9 Global versus local Gibbs-preserving convertibility

One of the central operational questions of the project is how the convertibility structure changes when one passes from arbitrary global Gibbs-preserving maps to tensor-product local Gibbs-preserving maps. The benchmark convertibility graph provides a clear answer: the local theory is dramatically more restrictive.

For the chosen set of 8 benchmark LT states, the adjacency counts were

$$\text{global edges} = 7, \quad \text{local verified edges} = 1, \quad \text{PPT-relaxation edges} = 1.$$

Thus, even on a very small benchmark set, most of the convertibility visible at the global level disappears once one imposes locality. This is exactly the qualitative phenomenon the project set out to detect.

At the benchmark-pair level, the TFD-to-dephased transition gives the cleanest positive example. The conversion is feasible under both global and verified local GP:

$$\text{Global GP} : \text{True}, \quad \text{Local GP (verified)} : \text{True},$$

with residuals well below tolerance. This is consistent with the analytic construction using local dephasing. Quantitatively, the conversion reduces the LT resource by exactly a factor of two:

$$I(\text{TFD}) = D(\text{TFD}) = 1.16441 \quad \longrightarrow \quad I(\text{deph}) = D(\text{deph}) = 0.582203.$$

Hence this family gives a concrete example of a strict monotone decrease that is achievable even under local GP.

By contrast, for a generic random LT pair both the global and local convertibility tests were negative, and the PPT outer relaxation was also infeasible. This supports the claim that incomparability is common, not exceptional, once one moves away from specially constructed families.

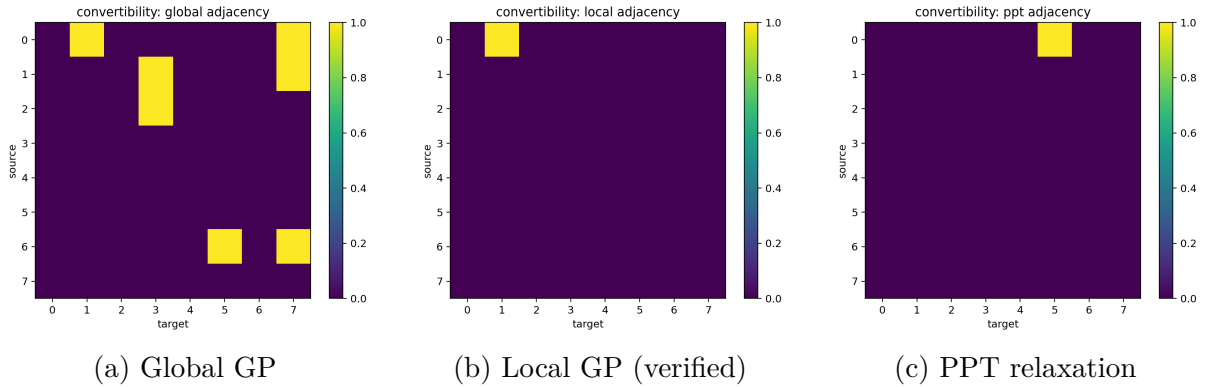


Figure 3: Convertibility adjacency matrices on a benchmark set of eight LT states. The global graph contains seven directed edges, whereas the verified local graph contains only one. This shows that locality imposes a severe additional constraint beyond Gibbs preservation alone.

10.10 Detailed convertibility structure on the Pauli- XX LT ray

The Pauli- XX LT ray provides a much more resolved picture than the small benchmark graph because it samples a one-parameter family across the full feasible interval

$$p \in [-0.7864477, 0.7864477].$$

Along this ray, the resource measures behave exactly as expected from the analytic theory. Both D and I are symmetric in p , attain their minimum at $p = 0$, and increase monotonically with $|p|$. The same qualitative behaviour is seen in the norms of the centered correlation operator $C = \rho - \Gamma$: both $\frac{1}{2}\|C\|_1$ and $\|C\|_F$ increase with $|p|$ and vanish at the thermal product point.

The convertibility graphs on this ray reveal a sharp separation between global and local GP.

For global GP, the adjacency count is

$$174,$$

and the corresponding heatmap exhibits a broad hourglass pattern centred at the least correlated states. Moreover, the reported number of monotone violations among global edges is

$$0.$$

Thus, within this family, global convertibility is strongly aligned with decreasing correlation strength: states at larger $|p|$ can typically be mapped to states closer to the centre, while the reverse directions are not seen.

For verified local GP, the adjacency count collapses to

$$20.$$

The heatmap is not hourglass-shaped; instead it is concentrated on the anti-diagonal, corresponding precisely to the pairing

$$p \longleftrightarrow -p.$$

This is a very informative result. It shows that along the XX ray, verified local convertibility does not generically contract the correlation magnitude. Rather, it mainly identifies states related by a local symmetry that flips the sign of the XX term while preserving both Gibbs marginals. This is exactly what one expects from diagonal local unitaries

such as a local phase flip:

$$(\sigma_z \otimes \mathbb{I})(\sigma_x \otimes \sigma_x)(\sigma_z \otimes \mathbb{I}) = -\sigma_x \otimes \sigma_x.$$

Hence the local graph is dominated by symmetry equivalences, not by a nontrivial local thermalisation order.

The PPT outer-relaxation graph lies strictly between the global and verified local graphs, with

67

directed edges. This is the expected qualitative behaviour for an outer bound: it is much less restrictive than verified local GP, but much more restrictive than unrestricted global GP. In the report, this should be described as evidence that PPT-type relaxations can capture some coarse structure of local convertibility, but are far from exact.

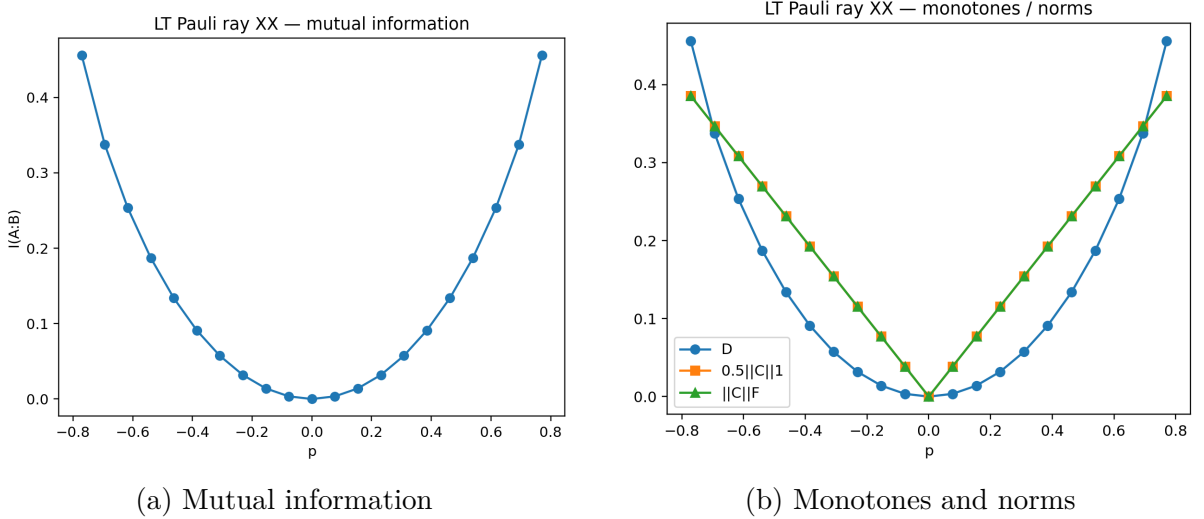


Figure 4: Resource profile along the Pauli-XX LT ray. The quantities D and I coincide, are symmetric in p , and are minimised at the product Gibbs point $p = 0$. The norms of the centered correlation operator show the same monotone growth with $|p|$.

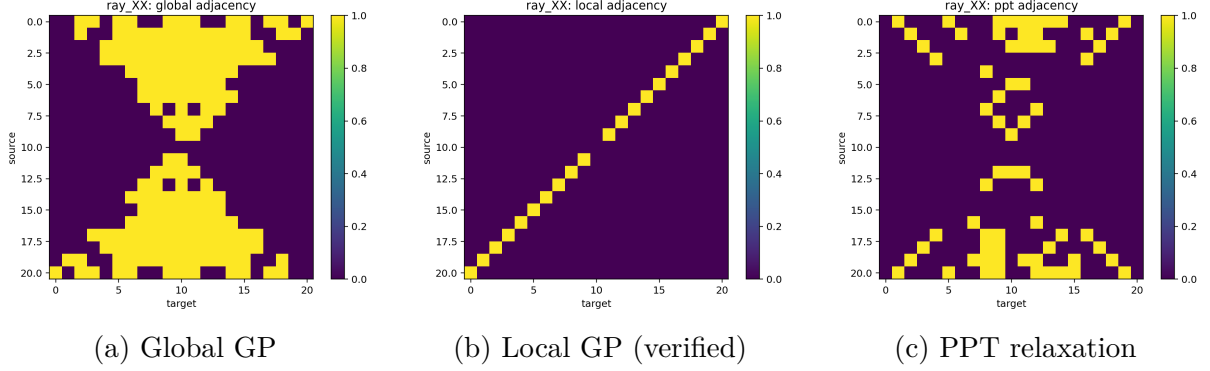


Figure 5: Convertibility structure on the Pauli-XX LT ray. Global GP admits many inward-directed conversions toward less correlated states, whereas verified local GP collapses mainly to sign-flip equivalences $p \leftrightarrow -p$. The PPT relaxation is intermediate and therefore useful as a coarse but non-exact outer approximation.

10.11 Mixing and dephasing paths from the TFD state

Two distinguished LT paths were studied numerically. The first is the dephasing path from the TFD state to its dephased version. The second is the mixing path from the TFD state to the product Gibbs state.

For the dephasing path, the endpoint conversion is feasible under both global and local GP, as discussed above. This is the cleanest numerical realisation of the analytic theorem that the TFD-to-dephased family is locally Gibbs-preserving convertible in the correlation-decreasing direction.

For the mixing path,

$$\rho_\lambda = (1 - \lambda) \tilde{\gamma} + \lambda \Gamma, \quad 0 \leq \lambda \leq 1,$$

the plot shows a smooth monotone decay of both D and I from the TFD value down to 0 at Γ . Since both endpoints and all convex mixtures remain locally thermal, this gives an explicit LT thermalisation path within the affine LT manifold itself. The curve is strictly decreasing and visually convex, with the steepest decay near the TFD end. This is consistent with the interpretation that highly correlated LT states lose resource rapidly once mixed with equilibrium, while states already near Γ have very little resource left to lose.

10.12 Geometry of the LT body

The boundary/interior geometry plot is only a low-dimensional projection of a high-dimensional convex body, so it should not be over-interpreted. Nevertheless, it captures the right qualitative picture. The interior points form a compact central cloud, while the boundary points wrap around them and exhibit a much larger spread in the projected

coordinates. This is exactly what one expects from a convex spectrahedral body: the interior is volumetric and relatively featureless under projection, while extremal and near-extremal points populate the visible shell.

This figure is therefore best used as a geometric illustration rather than as a theorem-producing object. Its role in the report is to visually support the analytic statement

$$\text{LT} = (\Gamma + \mathcal{Z}) \cap \text{PSD},$$

and to show that the product Gibbs state sits naturally near the centre of the accessible LT region, while more strongly correlated states are pushed toward the boundary.

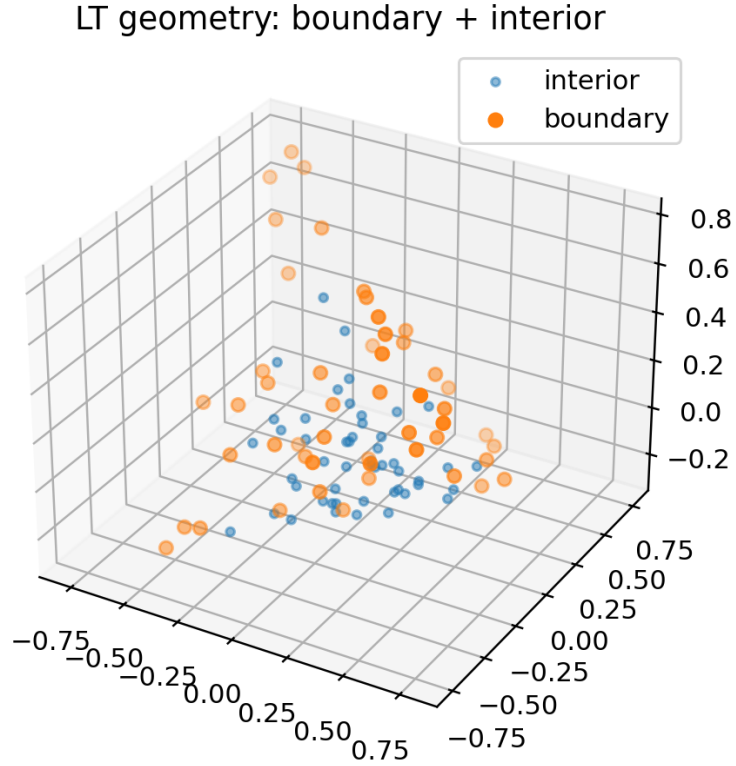


Figure 6: Low-dimensional embedding of sampled interior and boundary points of the two-qubit LT body. The figure should be interpreted as a projection of a high-dimensional convex spectrahedron rather than as a literal geometric cross-section.

10.13 Commuting qutrit LT subclass

The $d = 3$ commuting LT subclass behaves exactly like a classical equal-marginal transport family. In the sample of 250 states, the mutual information distribution is concentrated near zero, with empirical mean and standard deviation

$$\bar{I} = 0.0635303, \quad \text{std}(I) = 0.0595621.$$

The histogram has a strong low-correlation peak and a long right tail extending to about 0.33, showing that highly correlated commuting couplings do occur but are atypical under the present sampling rule.

As expected for a commuting family, the relative entropy to Γ again agrees with mutual information sample-by-sample. The reported mean minimum partial-transpose eigenvalue is positive,

$$\overline{\lambda_{\min}(\rho^{T_B})} = 0.00592668,$$

which is consistent with the fact that commuting states are diagonal in the product basis and therefore separable. This qutrit subsection is important because it gives a nontrivial higher-dimensional extension of the project without claiming a full 64-dimensional analytic classification. The numerics support the analytic transport-polytope picture: most random couplings are only weakly correlated, but the full commuting LT face still contains a substantial range of nonzero correlations.

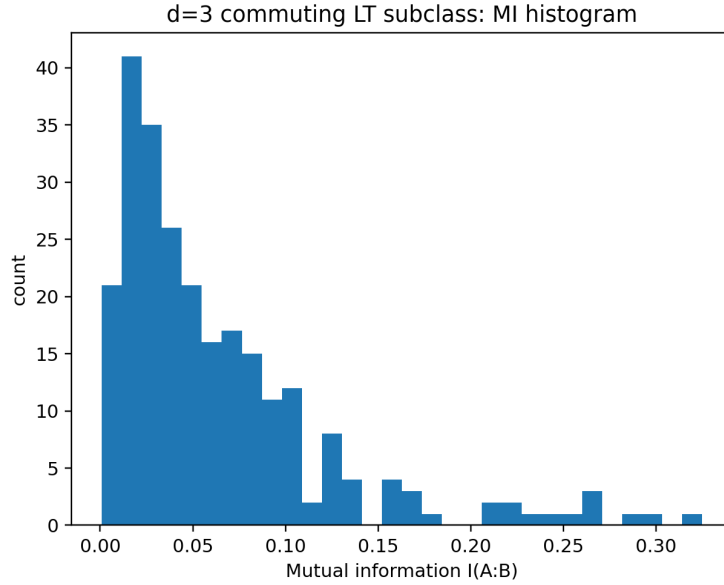


Figure 7: Histogram of mutual information in the sampled $d = 3$ commuting LT subclass. The distribution is strongly concentrated near low correlation, with a visible long right tail. This is consistent with the transport-polytope interpretation: strongly correlated commuting couplings exist, but are relatively atypical under random sampling.

10.14 Important numerical caveats

Two reported outputs should be treated as debugging targets rather than final physical conclusions.

First, the result

LT closure under random local GP channels: pass fraction = 0.033

is inconsistent with the analytic theorem that LT is closed under local Gibbs-preserving maps. If the sampled maps are genuinely local and Gibbs-preserving, then the pass fraction should be 1 up to numerical tolerance. Therefore at least one of the following must be wrong: the random channel generator, the Gibbs-preservation check, the LT-membership test after applying the channel, or the tolerance logic. This output should not be included as a scientific conclusion in the final report.

Second, if the PPT-on-joint-Choi test is being used as an outer relaxation of local GP, then any truly verified local edge should also appear feasible in the outer relaxation, up to numerical tolerance and identical indexing. If this inclusion fails in a plotted graph, then the issue is likely not physics but implementation: mismatched vertex order, inconsistent thresholds, or a bug in how the adjacency matrices are assembled. That should also be audited before the final write-up.

11 Discussion

11.1 What has actually been achieved

The project has achieved more than a collection of examples. Its core contribution is structural.

1. **Exact geometric formulation.** LT has been identified as a spectrahedron: an affine slice of the PSD cone.
2. **Main low-dimensional characterization.** In the qubit case, the full LT problem has been reduced to an explicit 9-parameter family and, generically, to concrete Schur-complement inequalities.
3. **Warmup subfamilies with exact tests.** The diagonal- T slice and LT rays admit exact positivity bounds, and the diagonal- T family admits an exact separability criterion.
4. **Thermodynamic meaning.** On LT, the free-energy monotone relative to equilibrium is exactly the mutual information.
5. **Operational structure.** Local GP maps preserve LT, and the TFD-to-dephased family provides an explicit ordered chain under local GP.
6. **Higher-dimensional extension.** In the qutrit case, the commuting LT subclass gives a rigorous and meaningful higher-dimensional analytic subproject with exact geometry and exact correlation bounds.

11.2 Why the project is genuinely nontrivial

The research question is not answered by standard textbook results. Fixing both marginals to thermal states imposes strong linear constraints, but positivity still leaves a highly nontrivial correlated body. Even in the qubit case the full LT set is 9-dimensional. Standard thermodynamic monotones such as free energy become informative but not complete, especially once one distinguishes global from local Gibbs-preserving maps.

The project is therefore not a routine application of known formulas; it sits exactly at the interface of:

- convex geometry of constrained quantum states,
- thermodynamic resource theory,

- correlation structure and entanglement,
- semidefinite-program-based convertibility.

11.3 Limitations and future work

The main analytic limitation is clear: while the full two-qubit LT problem is still tractable enough to study directly, the full qutrit LT problem is already too large for a comparably explicit treatment. This is why the commuting qutrit subclass is the right qutrit-level contribution.

The main next steps are:

1. fully polish the two-qubit Schur-complement theorem and, if space allows, relegate the $\Delta = 0$ boundary algebra to an appendix;
2. complete the numerical convertibility analysis with verified local edges only;
3. compare verified local convertibility with PPT outer relaxations;
4. investigate whether operator-Schmidt singular values of centered correlations provide useful necessary conditions for local convertibility;
5. explore classical reductions of commuting qutrit LT under global and local GP maps.

12 Conclusion

This project studies locally thermal bipartite states as a setting where all local athermality has been removed and any remaining thermodynamic resource is necessarily global. The resulting LT set is convex, structured, and operationally meaningful. On this manifold, relative entropy to product thermal equilibrium is exactly mutual information, so correlations acquire a direct thermodynamic interpretation.

The main analytic outcome is the near-complete treatment of the two-qubit problem. The full two-qubit LT set is a 9-dimensional spectrahedron, and its generic positivity can be reduced to explicit Schur-complement inequalities. This goes significantly beyond analysing a few special slices and provides the main structural theorem of the project. Supporting this, the diagonal- T family and ray constructions supply exact warmup analyses, including positivity, separability, and monotonicity tests. In higher dimension, the commuting qutrit LT subclass gives a tractable and nontrivial analytic extension with exact polytope structure and exact mutual-information range.

On the operational side, the report distinguishes global and local Gibbs-preserving convertibility, proves that LT is closed under local GP maps, and exhibits an explicit totally ordered TFD-to-dephased chain under local GP. Together with the SDPs implemented in the accompanying code, this provides a coherent framework for analysing both geometry and dynamics of locally thermal states.

A Appendix: concise proof of the LT dimension formula

A Hermitian operator on a $d_A d_B$ dimensional Hilbert space has $(d_A d_B)^2$ real parameters, and the trace-one condition removes one, leaving $(d_A d_B)^2 - 1$. Fixing the A marginal specifies a Hermitian trace-one matrix on $\mathcal{H}_A$, which imposes $d_A^2 - 1$ independent real conditions. Likewise fixing the B marginal imposes $d_B^2 - 1$ conditions. Subtracting gives

$$(d_A d_B)^2 - 1 - (d_A^2 - 1) - (d_B^2 - 1) = (d_A^2 - 1)(d_B^2 - 1).$$

B Appendix: notation for the main qubit formulas

For convenience, the central qubit quantities used in the report are collected here:

$$\gamma = \text{diag}(g, 1 - g), \quad \Gamma = \gamma \otimes \gamma, \quad \alpha = g - a, \quad \delta = 1 - g - a,$$

and in the gauge-fixed normal form

$$\rho = \begin{pmatrix} \alpha & x & y & z \\ x & a & w & -y \\ y & w^* & a & -x \\ z^* & -y & -x & \delta \end{pmatrix}, \quad x, y \in \mathbb{R}_{\geq 0}, \quad z, w \in \mathbb{C}.$$

The principal Schur parameter is

$$\Delta = a\alpha - x^2.$$

C Appendix: list of numerical figures to insert

At minimum, the final report should aim to include the following figure set:

1. LT geometry / embedding.

2. Diagonal- T family validation.
3. Convertibility graph: global versus local GP.
4. PPT outer relaxation for local GP.
5. TFD-to-dephased family.
6. Separable versus entangled LT.

Optional:

1. commuting qutrit LT sampling;
2. boundary scans or projection-to-LT diagnostics;
3. phase diagrams as a function of β if available.

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